```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
df = pd.read_csv('customer_churn.csv')
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Con
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Mo-mo
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	No	No	No	No	One year
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	Yes	No	No	No	Mo-mo
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Mo-mo

5 rows × 21 columns

replacing blanks with 0 as tenure is 0 and no total charges are recorded

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1 MB
```

```
df[df['TotalCharges'] == ' ']['TotalCharges'].replace(' ','0',0)
df['TotalCharges'] = df['TotalCharges'].astype('float')
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
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 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   float64
 20  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1 MB
```

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.describe
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2268.79897
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

```
df['SeniorCitizen'].replace({0: 'No', 1: 'Yes'}, inplace=True)
```

```
np.int64(0)
```

converted 0 and 1 value of senior citizen to yes/no to make it easier to understand

```
def convertedsc:
    if scval == 1:
        return 'Yes'
    else:
        return 'No'
df['SeniorCitizen'] = df['SeniorCitizen'].apply(conv)
```

```
df.head()
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Con
0	7590-VHVEG	Female	No	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Mo-mo
1	5575-GNVDE	Male	No	No	No	34	Yes	No	DSL	Yes	...	No	No	No	No	One year
2	3668-QPYBK	Male	No	No	No	2	Yes	No	DSL	Yes	...	Yes	No	No	No	Mo-mo
3	7795-CFOCW	Male	No	No	No	45	No	No phone service	DSL	Yes	...	Yes	Yes	No	No	One year
4	9237-HQITU	Female	No	No	No	2	Yes	No	Fiber optic	No	...	No	No	No	No	Mo-mo
5	9305-CDSKC	Female	No	No	No	8	Yes	Yes	Fiber optic	No	...	Yes	No	Yes	Yes	Mo-mo
6	1452-KIOVK	Male	No	No	Yes	22	Yes	Yes	Fiber optic	No	...	No	No	Yes	Yes	Mo-mo
7	6713-OKOMC	Female	No	No	No	10	No	No phone service	DSL	Yes	...	No	No	No	No	Mo-mo
8	7892-POOKP	Female	No	Yes	No	28	Yes	Yes	Fiber optic	No	...	Yes	Yes	Yes	Yes	Mo-mo
9	6388-TABGU	Male	No	No	Yes	62	Yes	No	DSL	Yes	...	No	No	Yes	No	One year
10	9763-GRSKD	Male	No	Yes	Yes	13	Yes	No	DSL	Yes	...	No	Yes	No	No	Mo-mo
11	7469-LKBCI	Male	No	No	No	16	Yes	No	No internet service	No internet service	...	No internet service	No internet service	No internet service	No internet service	Mo-mo
12	8091-TDKZJ	Male	No	Yes	No	58	Yes	Yes	Fiber optic	No	...	No	No	Yes	Yes	One year
13	0280-XJGEX	Male	No	No	No	49	Yes	Yes	Fiber optic	No	...	Yes	Yes	Yes	Yes	Mo-mo
14	5129-JLPIS	Male	No	No	No	25	Yes	No	Fiber optic	Yes	...	Yes	No	Yes	Yes	Mo-mo
15	3655-SNQYZ	Female	No	Yes	Yes	69	Yes	Yes	Fiber optic	Yes	...	Yes	Yes	Yes	Yes	Two year
16	8191-XAZND	Female	No	No	No	52	Yes	No	No	No internet service	...	No internet service	No internet service	No internet service	No internet service	Mo-mo
17	9959-WOFKT	Male	No	No	Yes	71	Yes	Yes	Fiber optic	No	...	Yes	No	Yes	Yes	One year
18	4190-MFLUW	Female	No	Yes	Yes	10	Yes	No	DSL	No	...	No	No	No	No	Mo-mo
19	4703-QXPVQ	Female	No	No	No	21	Yes	No	Fiber optic	No	...	No	No	Yes	Yes	Mo-mo
20	8779-QRDMV	Male	No	yes	No	1	No	No phone service	DSL	No	...	No	No	No	No	Mo-mo
21	1680-VDCWW	Male	No	Yes	No	12	Yes	No	No	No internet service	...	No internet service	No internet service	No internet service	No internet service	Mo-mo

22 rows × 21 columns

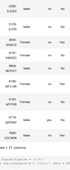
```
plt.figure(figsize = (5,4))
ax = sns.countplot x = 'Churn', data = df)
ax.bar_label(ax.containers[0])
plt.title('count of customers by Churn')
plt.show()
```

From the give pie chart we can conclude that 26.54% of our customers have churned out.

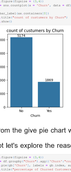
not let's explore the reason behind it.

```
plt.figure(figsize = (4,4))
df = df.groupby('Churn').agg('Churn',"count")
plt.pie(df['Churn'], labels = df.index, autopct = '%1.1f%')
plt.title('percentage of Churned Customers')
plt.show()
```

```
plt.figure(figsize = (4,4))
ax = sns.countplot x = 'gender', data = df, hue = 'Churn')
ax.bar_label(ax.containers)
plt.title('Churn By Gender')
plt.show()
```

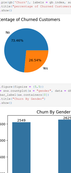



```
plt.figure(figsize = (4,4))
ax = sns.countplot x = 'SeniorCitizen', data = df)
ax.bar_label(ax.containers)
plt.title('Count of customers by Senior Citizen')
plt.show()
```



comparative a greated percentage of people is senior citizen category have chured.

```
# Senior citizen and churn
df = pd.crosstab(df['SeniorCitizen'], df['Churn'])
# Calculate percentages
df = df / df.sum(axis=1) * 100
# Plot stacked bar chart
ax = df.plot.bar(stack=True, color=['#ff6b6b', '#4ecdc4'])
# Add labels
for c in ax.containers:
    ax.bar_label(c, fmt='%.1f%%', label_type='center')
plt.title('Churn By Senior Citizen (%)')
plt.ylabel('Percentage')
plt.legend(title='Churn')
plt.show()
```



people who have use our services for a long time have stayed and people who have used our services 1 or 2 month have churned

```
plt.figure(figsize = (6,4))
ax = sns.histplot x = 'tenure', data = df, hue = 'Churn', bins = 72)
plt.show()
```

people who have month Contract are likely to churn then from those who have 1 to 2 years of Contract.

```
plt.figure(figsize = (6,4))
ax = sns.countplot x = 'Contract', data = df, hue = 'Churn')
ax.bar_label(ax.containers)
plt.title('Count of customers by Contract')
plt.show()
```

```
df.columns.values
```

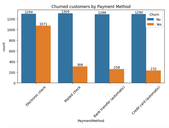
```
array(['customerID', 'gender', 'SeniorCitizen',
       'Partner', 'Dependents', 'tenure',
       'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
       'TechSupport', 'StreamingTV', 'StreamingMovies',
       'Contract', 'PaperlessBilling', 'PaymentMethod',
       'MonthlyCharges', 'TotalCharges', 'Churn',
       dtype=object)
```

Customers with Fiber optic internet show noticeably higher churn compared to DSL users, indicating possible dissatisfaction with pricing or service quality. Users without OnlineSecurity, TechSupport, or DeviceProtection churn more frequently, suggesting these add-on services improve retention. Customers who do not have internet service generally show very low churn. Overall, having multiple service features (security, backup, support) appears strongly associated with reduced churn risk. 📊

```
# Create visualization with subplots for all features
cols = [
    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies'
]

# Create figure and axes
fig, axes = plt.subplots(3, 3, figsize=(15, 12))

# Flatten axes for easy iteration
axes = axes.flatten()

for i, col in enumerate(cols):
    ax = sns.countplot(x=col, hue='Churn', data=df, ax=axes[i], legend=i == 0)
    ax.set_title(f'{col} by Churn')
    ax.set_ylabel('Count')
    ax.legend(title='Churn')

plt.tight_layout()
plt.show()
```

customer is likely to churn he is using electronic check as a payment method.

```
plt.figure(figsize = (6,4))
ax = sns.countplot x = 'PaymentMethod', data = df, hue = 'Churn')
ax.bar_label(ax.containers)
ax.set_xticklabels(ax.get_xticklabels())
plt.title('Churns Customers by Payment Method')
plt.legend(title='Churn')
plt.show()
```


Churned customers by Payment Method
Churn
No
Yes
1294
1071
1304
308
1308
274
1286
232
count
0
200
400
600
800
1000
1200
Electronic check
Mailed check
Bank transfer (automatic)
Credit card (automatic)
PaymentMethod